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PALAEOMAGNETIC DATA FOR PERMIAN AND TRIASSIC ROCKS FROM DRILL HOLES IN THE SOUTHERN SYDNEY BASIN, NEW SOUTH WALES

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ABSTRACT

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A section 300 m thick across the Permian—Triassic boundary has been sampled in the Southern Coalfield of the Sydney Basin, New South Wales. 55 samples, mainly grey to drab sandstones, were collected from 9 diamond drill holes which penetrated the entire Narrabeen Group and the upper part of the conformably underlying Illawarra Coal Measures, as well as a sill emplaced into the coal measures. The samples included fully oriented cores. Additional reconnaissance samples from two further drill holes were also studied.

Partial alternating field demagnetization and petrography indicate the magnetic remanence to be a stable DRM. Partial thermal demagnetization above 300°C or 400°C caused large increases in magnetic susceptibility. Partial chemical demagnetization did not cause significant changes in remanence directions.

For the Coal Cliff Sandstone (basal Narrabeen Group, Triassic) the palaeomagnetic pole position (Normal) was calculated to be at 59°N 322°E (dp = 27°, dm = 29°), which agrees with previously published data. For the uppermost coal measures (Permian) the pole position was calculated as 58°N 340°E (dp = 09°, dm = 10°). Data for samples from the lower to middle coal measures yield a pole position which is between the new Permian—Triassic pole position and that for the underlying Middle Permian igneous rocks. The top of the Reversed "Kiaman Magnetic Interval" (Permian) may be near the Tongarra coal and Appin Formation boundary — (early) Late Permian.

INTRODUCTION

Mercanton (1926) reported reversed remanent magnetization in Permian basaltic rocks ("latites") from the Shoalhaven Group (Table I) of the southern Sydney Basin, New South Wales. Irving and Parry (1963) and Irving (1963) studied the palaeomagnetism of Permian and Triassic rocks from the southern Sydney Basin, but their results did not span the section across the top of the Permian (Illawarra Coal Measures) and the base of the Triassic (Narrabeen Group), which is approximately 370 m thick near Wollongong

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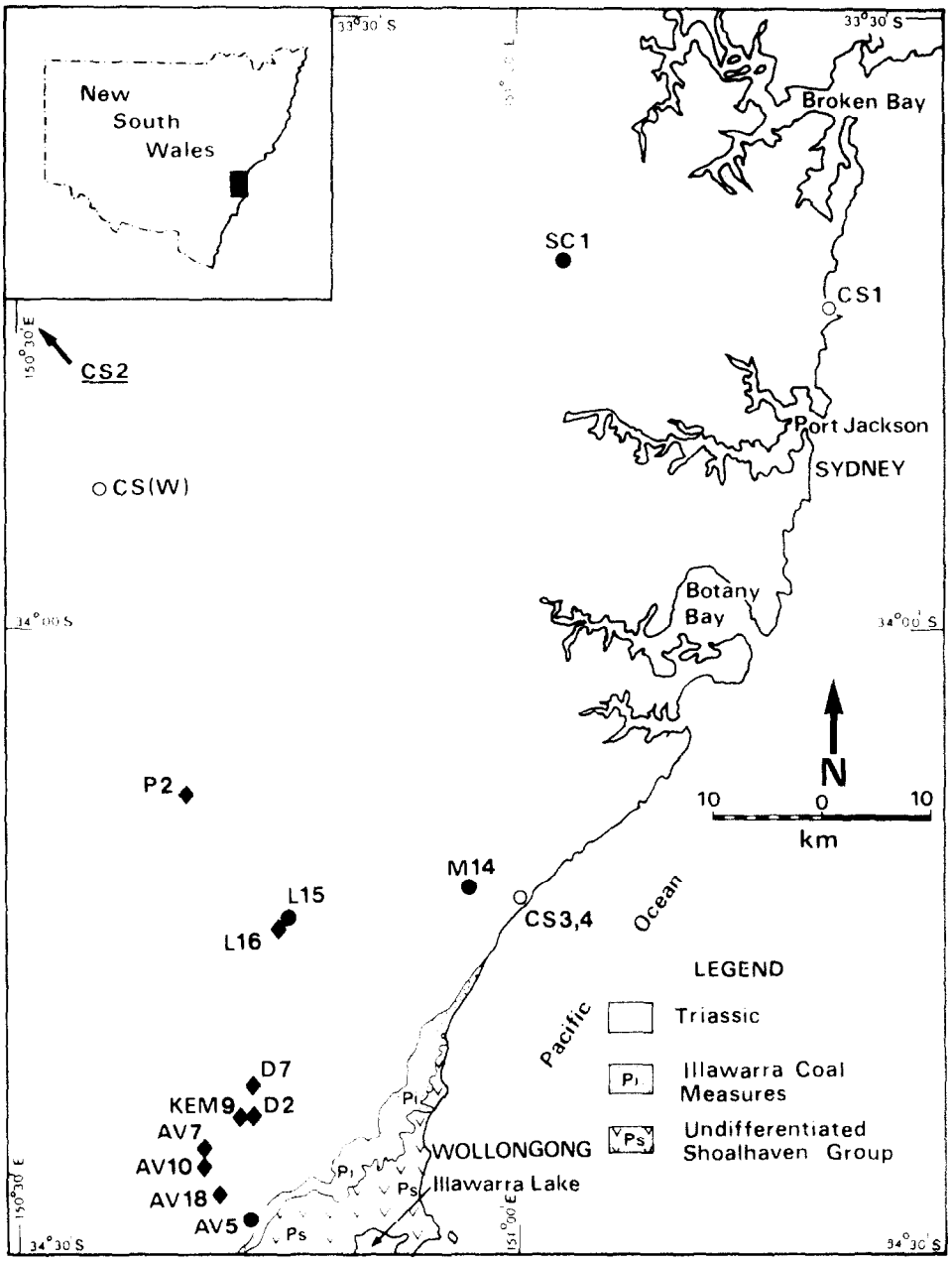


Fig. 1. Locality map showing the sampling sites for the material presented and discussed here, based in fig. 1 of Facer (1973) (the drill holes studied for that investigation being represented by solid dots). Irving's (1963) "chocolate shales" (CS) sites are shown (CS(W)) being his Warragamba Dam site (CS2 is at 33°35'S 150°14'E). The Mt. Gibraltar and Gingenbullen localities of Boesen et al. (1961) are at approximately 34°25'S 150°23'E. The latite samples (Irving and Parry, 1963) came from the Shoalhaven Group from the area shown here and extending to the south; and the monzonite at Milton (Robertson, 1964) is at 35°20'S 150°25'E. (AV is an abbreviation for Avon; D for Dendrobium; KEM for Kemira; L for Loddon; M for Metropolitan; P for Picton; and SC for South Colah.) Drill hole Wollongong 35 is at 34°16'S 150°42'E and drill hole Huntley 4 at 34°34'S 150°37'E.

TABLE I

Stratigraphic sequence, Southern Coalfield, southern Sydney Basin, New South Wales *

Age	Unit	Thickness [c] (m)
Post-Triassic	Various intrusions and extrusions	—
Triassic	Wianamatta Group	thin remnants
	Hawkesbury Sandstone	~150
	Narrabeen Group	
	Gosford Sandstone [a]	11
	Clifton Sub-Group	
	Bald Hill Claystone [a]	15
	Bulgo Sandstone	120
	Stanwell Park Claystone [a]	37
	Scarborough Sandstone [a]	26
	Wombarra Shale [a]	37
	Coal Cliff Sandstone [a]	9
Permian	Illawarra Coal Measures	
	Bulli seam	2
	Sandstone [a] and shale [a]	15
	Balgownie seam	1
	Sandstone [a], shale, coal	15
	Wongawilli seam [b]	10
	Kembla Formation [a]	15
	Sandstone, shale, coal (including Allans Creek and Appin Formations, Tongarra coal) [a]	60
	Shoalhaven Group	~1100
	(Clyde Coal Measures	~40)

* Data from tables in Packham (1969). Some sandstone horizons include conglomerate. Shale and claystone both occur in the fine-grained horizons.

[a] Sampled unit.

[b] The sill sampled from within the Wongawilli seam.

[c] Thicknesses are approximate because of significant variation throughout the Southern Coalfield, the thicknesses shown being those corresponding to the mean position of the localities sampled here (taken to be 34.4°S 150.7°E). The Illawarra Coal Measures and Narrabeen Group thicken towards the north.

(Fig. 1 and Table I). The Permian to Mesozoic sedimentary rocks of the southern Sydney Basin can thus provide palaeomagnetic data to extend discussion of the *Kiaman Magnetic Interval* (Irving and Parry, 1963).

Extensive descriptions of the geology of the southern Sydney Basin have been presented by various authors (e.g. Packham, 1969; and Mayne et al., 1974). Apparently sedimentation commenced during the Late Carboniferous with freshwater sedimentation, with deposition during the Permian including the marine sedimentary rocks and extensive volcanic rocks of the Shoalhaven Group, which in turn passed upwards into thick freshwater sequences

of the Late Permian Illawarra Coal Measures (Wilson, 1969). The coal measures sedimentation is conformable with the overlying Triassic Narrabeen Group of sandstones and shales or claystones with, in the Southern Coalfield (southeastern Sydney Basin), thin cappings of Hawkesbury Sandstone and isolated remnants of Wianamatta Group (shales) (Table I).

Exploitation of coal and latite has prompted studies to correlate the Permian rocks across the Southern Coalfield, although some problematic aspects of the correlation remain. Facer (1973) showed that magnetic remanence of bore cores presented hope of providing useful time horizons. Further sampling and investigation have since been carried out. (Due to instrument calibration problems "Normal" should be "Reversed" and vice versa in fig. 2 of Facer (1973), although the conclusions remain valid.)

PREVIOUS PALAEOMAGNETIC STUDIES OF THE SOUTHERN SYDNEY BASIN

Reversed magnetization was detected by Mercanton (1926) in Permian (Shoalhaven Group) latite flows from Kiama (about 30 km south of Wollongong) and later established by the work of Irving and Green (1958) and Irving and Parry (1963). Irving (1963) found that surface exposures of red claystones (informally known as "chocolate shales") of the Narrabeen Group carried normal magnetization. The reversed period of magnetization is now often termed the *Kiaman Magnetic Interval*, although Irving (1971, p. 530) suggested "that the term 'late Palaeozoic Reversed Interval' be used".

The palaeomagnetism of igneous rocks intrusive into the Sydney Basin sedimentary rocks has been studied by Boesen et al. (1961) and Schmidt (1976) (Jurassic rocks) and Robertson (1964) (a Permian intrusion). Radiometric ages for these rocks have been recalculated by Facer and Carr (1979) using new decay constants of Steiger and Jäger (1977) (cf. Table IV). However, recent discussion (e.g. Green, 1977) of Mesozoic palaeomagnetic data and the apparent polar wander path (at least partly based on data from Mesozoic rocks of the Sydney Basin) suggests that additional work may be necessary.

Agostini (1963) and Henderson (1964) carried out palaeomagnetic studies on igneous rocks in the Wollongong area to complement ground magnetic surveys. Surface samples apparently possessed components of secondary magnetization which could not be effectively removed (Henderson, 1964). Inclinations in bore core samples were variable, even within individual igneous bodies (Agostini, 1963).

TECHNIQUES

Field measurements and sampling

The magnetic "logging" technique described by Facer (1973) was used to study the NRM of drill core from Australian Iron and Steel (AIS) vertical

diamond drill hole AV7 (Table II and Fig. 1). 38 samples were selected to provide as wide a range of rock types (but mainly grey to drab sandstones) and stratigraphic units as possible. These samples are partially oriented, with only the vertical (probably within 3°) axis and upward direction known. Some shale or claystone horizons in the Narrabeen Group were too fragmented for sampling of coherent core. Additional samples were collected from cores of 8 other vertical drill holes (Fig. 1 and Table II). Of these additional samples 11 (from 5 drill holes) were fully-oriented cores, 5 (AIS D2) were partially-oriented cores from a thick sill; and one sample from AIS AV5 was of a dense claystone which apparently possessed a reversal.

Specimens 25 mm in diameter and nominally 23 mm in length were prepared (Table II). Some cores, including most coarse-grained sandstones from the top of the Narrabeen Group, broke up during the specimen preparation. Thin sections, and if possible polished sections, were made of most samples.

An additional 17 samples, collected from two other drill holes (sampled for thermal studies) extended the area and the stratigraphic interval investi-

TABLE II

Sample and specimen preparation data

Hole identification	No. of samples	No. of specimens
Avon 5 (AV5)	1 [a]	5
Avon 7 (AV7)	38 [b]	55
Avon 10 (AV10)	4 [c]	9
Avon 18 (AV18)	2 [d]	7
Dendrobium 2 (D2)	5 [e]	10
Dendrobium 7 (D7)	1 [f]	2
Kemira 9 (KEM9)	2 [g]	5
Loddon 16 (L16)	1 [h]	2
Picton 2 (P2)	1 [i]	3

Notes: All diamond drill holes were drilled by Australian Iron and Steel Pty. Ltd. (Coal Geology). For stratigraphic details see Table I and Wilson (1969).

[a] The AV5 sample was from a depth of 256 m in the Appin Formation.

[b] AV7 samples span the section from the top of the Narrabeen Group to the Wongawilli seam.

[c] Oriented cores from the Coal Cliff Sandstone (158 m); between the Bulli and Balgownie seams (2)(186 m); and below the Wongawilli seam (218 m).

[d] Oriented cores from the Bulgo Sandstone (62 m) and between the Bulli and Balgownie seams (139 m).

[e] Samples (278 m to 303 m) from a 31 m thick sill.

[f] Oriented core between the Bulli and Balgownie seams (290 m).

[g] Oriented core between the Bulli and Balgownie seams (91 m).

[h] Oriented core between the Bulli and Balgownie seams (469 m).

[i] Oriented core from the base of the Coal Cliff Sandstone (depth 691 m).

The reconnaissance samples came from N.S.W. Department of Mines drill holes Wollongong 35 and Huntley 4 (for N.S.W. Electricity Commission).

gated, although these samples only provided complementary information such as demagnetization characteristics.

Magnetic remanence

The NRM's of all specimens were measured on either a "Digico" or a "Schonstedt" slow-speed spinner magnetometer.

Partial AF demagnetization was carried out in steps up to 100 mT (1000 oersted) in both a "Schonstedt" demagnetizer and a custom-built 3-axis tumbler demagnetizing instrument. As all igneous rock specimens in this investigation were subsequently used for a thermal conductivity study they were only subjected to partial AF demagnetization.

Partial thermal demagnetization and treatment up to 800°C was carried out on 13 specimens (and 12 reconnaissance specimens) using a "Schonstedt" demagnetizer. Heating and cooling were in air in field-free space (5 nT or less). Thin section studies had indicated that lack of red colouration in samples from depths greater than 65 m in AV7 (colours grey to drab) reflected almost complete lack of iron-oxide cement, although minor hydroxides were present.

Partial chemical demagnetization was also carried out. The specimens were partly "perforated" by a thin-walled 10 mm diameter drill bit to increase the potential reaction surface. Leaching was carried out in 8*N* HCl. After each step the specimens were washed, soaked, and then dried in field free space.

Magnetic susceptibility anisotropy

By courtesy of D.T.A. Symons and M. Stupavsky the magnetic susceptibility anisotropy of 16 specimens was determined (cf. Christie and Symons, 1969). These specimens represented all sedimentary rock types of this investigation. Some specimens were measured after partial thermal treatment (up to 650°C or 700°C).

Direction mean calculations

Mean directions of magnetization for fully oriented cores were calculated using the technique of Fisher (1953) (cf. Facer, 1976). The mean inclinations for groups of cores were also calculated arithmetically, which yielded mean inclinations about 2° shallower than those obtained from Fisherian analysis. However, for this investigation the means calculated using the two techniques would not be statistically distinguishable. All specimens were used in the mean calculations except as discussed in the text.

RESULTS

Field magnetic results

Magnetic "logging" of the AV7 core did not indicate definite reversals, although at several horizons within the Narrabeen Group indications of reversals were detected (the doubt arising because the intensity of the remanence was so low for many samples that the remanence could not be detected by the field instrument). The apparent reversal near the boundary of the Narrabeen Group and Hawkesbury Sandstone in AV5 (Facer, 1973, fig. 2) was not detected in AV7.

Laboratory measurements on AV7 samples detected several NRM reversals, and partial demagnetization did not change these observations. Table III presents NRM data, the intensities for all sedimentary rocks below the Bald Hill Claystone being low to very low. As no samples were fully oriented, core inversion during storage prior to sampling may have occurred and only limited significance can be attached to these (directional) results until further measurements are made. The apparent reversal within sample AV5 256 (sample from depth of 256 m in AV5) was confirmed. No reversals were detected *within* other samples. As AV18 62 and AV18 139 were fully oriented, their opposite polarity is not an artefact of storage. AV18 62 is grey, reflecting the absence of significant amounts of secondary iron oxides. Table IV contains data from the fully-oriented cores and from published results elsewhere in the area.

Partial alternating field demagnetization

Partial AF demagnetization in fields up to 20 mT of all specimens of the sill from D2 removed approximately 50% of the NRM. Beyond 20 mT AF demagnetization reduced the intensity significantly, with 10% or less remaining by 100 mT.

Behaviour of the sedimentary rocks during partial AF demagnetization differed between the reddened specimens from AV7 and the grey or drab specimens. For reddened specimens 20—25% of initial NRM intensity remained after demagnetization at 100 mT, whereas for the grey or drab specimens the intensity reached this level at only 25 mT to 30 mT. This difference in response to partial AF demagnetization is consistent with the observation that detrital magnetite is a significant carrier of remanence in the grey and drab specimens and that (secondary) hematite cement contributes to the remanence of the red specimens. Stability of reconaissance specimens supported these observations.

Partial AF demagnetization caused a slight steepening of inclinations (cf. Table III) up to 25 mT, after which the intensity was not sufficient for accurate assessment.

TABLE III

Palaeomagnetic results (partially oriented)

Unit, hole sampled		Number	J_n		I_n		I_{pd}	
			mean	s.d.	mean	s.d.	mean	s.d.
Bald Hill Claystone, AV7	n	2	10.6	1.8	−66.5	13.4	−69.0	09.9
	(n *	7	26.9	16.0	−71.9	11.3)		
Bulgo Sandstone, AV7, AV18	n	14	1.49	0.71	−74.0	06.4	−77.7	04.6
	(n *	24	1.24	0.66	−72.0	09.0)		
Stanwell Park Claystone, AV7	(n *	1	0.70	—	−87.0	—)		
Scarborough Sandstone, AV7	n	4	1.59	0.89	−77.0	01.8	−78.5	07.7
	(n *	12	1.61	0.51	−75.8	03.7)		
Coal Cliff Sandstone, AV7, AV10, P2	n	8	1.05	0.44	−72.9	05.2	−77.6	04.0
	H	3	1.10	0.34	−72.7	01.7	−77.3	
	(n *	11	0.95	0.41	−72.7	04.9)		
Between Bulli and Balgownie seams, AV7, AV10, AV18, D7, KEM9, L16	n	16	1.08	0.33	−69.7	08.7	−73.9	04.6
	H	6	1.05	0.38	−72.5	06.7	−74.7	04.4
	(n *	18	1.11	0.39	−70.6	08.5)		
Below Balgownie seam, AV7, AV10	n	6 [a]	0.78	0.27	−75.2	03.7	−77.3	06.6
	H	2	0.80	(0.24)	−74.7	(02.3)	−78.8	(02.2)
	(n *	7	0.77	0.25	−75.4	03.4)		
Sill (D2)	n	10	9.59	6.60	−75.5	03.7	−76.2	04.8

[a] In AV7 the lowest two specimens have very low intensity (and shallow inclination) and have been omitted.

Results for AV5 are not included. The upper number n refers to specimens; the number H, refers to holes, and the number n * refers to all specimens, including those not partially demagnetized. (Three specimens could not be measured.)

J_n is the NRM intensity in units of 10^{-3} A/m; I_n, I_{pd} are the inclinations in degrees of the NRM, and after partial AF demagnetization or after partial thermal demagnetization at 300°C to 350°C. All are assumed to be negative or normal for ease of comparison. Partial chemical demagnetization data are not included (see text). The AF selected in all cases was 15 or 20 mT, although higher fields were used to test stability more fully.

Results for hole numbers AV10, AV18, D7, KEM9, L16 and P2 are for fully-oriented cores (cf. Table IV), but are included here in order that all inclinations can be compared.

TABLE IV

Permian and Mesozoic palaeomagnetic directions in rocks from the Southern Coalfield, southern Sydney Basin [a]

Unit	Age	Palaeomagnetic data								Reference
		P	*N*	*n*	*D*	*I*	*R*	*k*	α_{95}	
Narrabeen Group	Triassic	N	5	(62)	—	−79	(s.d. 02.1)			Irving (1963)(recalc.)
		T	4	(32)	337	−82	3.97	85	10.0	
					or	−79	(s.d. 02.1)			
Narrabeen (Bulgo Sandstone)	Triassic	(A, T) *	(1)	3	332	+76	2.94	34	21.6	this study
Narrabeen (Coal Cliff Sandstone)	Triassic	A (T) *	2	(5)	349	−77	1.99	266	15.4	this study
Illawarra Coal Measures (upper)	Permian	A (T) *	3	(6)	011	−77	2.98	100	05.2	this study
Illawarra Coal Measures (middle)	Permian	A	(1)	4	189	−80	3.98	187	06.7	this study
Milton monzonite	Permian (245)	A	4	(18)	085	+81	—	39	15.7	Robertson (1964)
Shoalhaven Group (latites)	Permian (253)	A	11	(158)	222	+81	10.6	27	08.9	Irving and Parry (1963)(recalc.)
		or	10	(147)	205	+83	9.82	50	06.9	

[a] For localities see Fig. 1. The age is the stratigraphic age, with radiometric ages (new and recalculated, and plotted in Facer and Carr, 1979) in millions of years if known.

P is the partial demagnetization treatment applied to the specimens (A, AF; N, NRM; T, Thermal). * As noted in the text the results of partial thermal demagnetization of this investigation have not been used in these calculations. Insufficient data are available for partial chemical demagnetization results to be included here.

N is the number of samples or "sites", *n* the number of specimens, used in Fisher (1953) analysis, with the Fisherian statistics being for the number not in parenthesis.

D, *I* are the declination, inclination in degrees of the mean direction and *R*, *k* and α_{95} the various parameters of precision as calculated using Fisher (1953) and the computer program of Facer (1976). If $\alpha_{95} > 20^{\circ}$ the population is considered to be too scattered.

Partial thermal demagnetization

Twelve grey and drab specimens from 60 m or greater depth were treated by partial thermal demagnetization. In all cases, except for two specimens from AV5 which experienced major physical breakdown by 300°C, directions showed little change up to 400°C, with slight steepening of inclination. Above 400°C the directions changed rapidly and irregularly. The intensity also changed irregularly — but generally with a marked increase. The magnetic susceptibility increased strongly above 450°C and continued to increase up to 700°C or 750°C, after which some specimens showed a slight decrease to 800°C. For most specimens the ratio of susceptibility after heating at 700°C to that at 20°C is over 50, the most pronounced increase being a factor of over 1000 (cf. Table V). Nine reconnaissance specimens showed similar behaviour.

Changes in intensity and susceptibility were more regular in a specimen from AV7 64 which contains extensive red iron oxide cement, and two reconnaissance specimens from Huntley 4 from depths to 65 m. Although the intensity was much more constant than in the grey specimens, even above 600°C, the susceptibility increased sharply at 500°C. Changes in direction above 450°C were just as marked as in the deeper specimens. Due to difficulties in preparing specimens no untreated reddened specimens remained for confirmatory treatment.

Colour changes during heating of the grey and drab specimens included bleaching, then darkening, then bleaching again. External colour changes were generally less pronounced than the internal darkening, even to black,

TABLE V

Magnetic susceptibility anisotropy

Specimen no.	Magnetic suscept. ($\times 10^{-4}$)	Maximum anisotropy		Lineation $\left(\frac{\chi_{max}}{\chi_{int}}\right)$	Foliation $\left(\frac{\chi_{int.}}{\chi_{min.}}\right)$	Eccentricity $\left(\frac{\text{foliation}}{\text{lineation}}\right)$
		decl.	incl.			
AV7 214B	0.16	303°	+05°	1.0063	1.0109	1.005
AV7 225A	0.67	038°	+08°	1.0033	1.0124	1.009
AV7 225B(H)	159.0	052°	+06°	1.0028	1.0262	1.023
AV18 62A(H)	56.0	347°	+01°	1.0042	1.0259	1.022
AV18 139B(H)	7.7	329°	+03°	1.0041	1.0301	1.026

11 other specimens were insufficiently anisotropic to include in this table.
Note: (H) These specimens were heated during partial thermal demagnetization to 650°C. Four other specimens heated to 650°C or 700°C were insufficiently anisotropic for inclusion in this table.
Lineation: This ratio indicates the degree of anisotropy: 1.0063 indicates 0.63% anisotropy.

within some specimens at high temperatures. For the red specimen only bleaching was apparent.

All specimens could acquire a very soft viscous remanence (VRM) after heating above 400°C and thus considerable care was taken to minimize VRM acquisition during transfer from the shielded cooling chamber to the magnetometer. VRM's were remeasured several times — following storage in the ambient laboratory field, after periods varying from ten minutes up to several days after most heat treatments from 400°C. Groups of specimens were stored with common orientations, and directions between specimens agreed closely throughout the measurements even after storage inversion which caused rapid VRM "reversal" within minutes. VRM intensity was as much as 1000 × NRM intensity.

Partial chemical demagnetization

Some specimens broke down during the leaching treatment, especially the sideritic claystone from AV5. The only specimens to show noticeable colour changes were a red-coloured specimen (56 m depth, AV7) and two specimens which had previously been subjected to partial thermal treatment — all of which showed external bleaching (and reduced internal darkening if previously heated). Apart from the previously-heated specimens there was little change in either intensity or direction after partial chemical demagnetization of up to 145 hr. Leaching periods longer than 145 hr were precluded by disintegration.

Previously-heated specimens showed initial reduction in intensity to about one-third, but intensity doubled again following storage of up to 1.5 hr in the laboratory, indicating that the magnetic constituents capable of acquiring a VRM were still present. This test was repeated after various periods of acid treatment with the same result.

Because of limited directional changes during partial chemical demagnetization the results were used in a qualifying fashion only.

Magnetic susceptibility anisotropy

Table V contains details of the anisotropy of magnetic susceptibility. Only three specimens possessed sufficient anisotropy for results to be considered reliable (D.T.A. Symons, pers. commun., 1979), although another two provided reasonably useful data. Heating did not cause notable changes in anisotropy.

The weak susceptibility (unheated specimens) and low degree of anisotropy (Table V) indicates that no significant influence was exerted on magnetic directions by the anisotropy.

Reversal in AV5

The AV5 256 sample (14 cm length) was cut into 5 specimens. The rock is a sideritic claystone, buff to grey when fresh (with no noticeable discoloura-

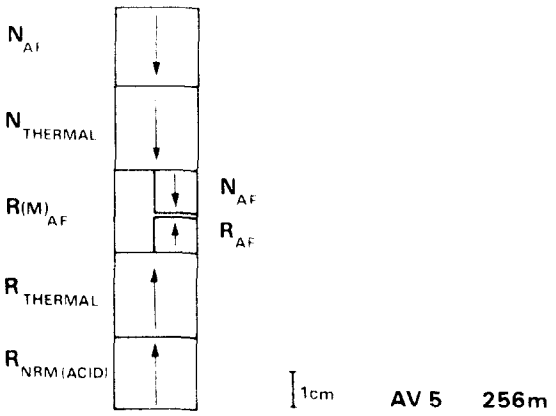


Fig. 2. Schematic representation of the reversal in the sample from a depth of 256 m in AV5. The polarity is indicated by N ("Normal"), R ("Reversed") and M ("Mixed"), with the partial demagnetization treatment as a subscript. The inclination direction is indicated by the arrows. Details are given in the text.

tion during 6 months of storage after specimen preparation). Thin section study indicated a very fine grain size — fine to very fine clay and siderite particles — which rendered mineralogical identification most difficult.

Of the specimens, the top 2 were "Normal" and the lower 3 "Reversed" (intensities down to 10^{-4} A/m or less). Figure 2 schematically presents inclination data after partial demagnetization. The top specimen possessed slight hematite staining along a calcite-filled fracture 0.5 mm thick — and possessed reasonable stability to partial AF demagnetization to 100 mT. No significant magnetic susceptibility anisotropy was detected. The middle specimen possessed very low remanence which declined quickly during partial AF demagnetization to 25 mT. Hence the specimen was sawn in two and AF treatment continued to 100 mT. Although the moment remained low, each half was more easily measured, the upper half being "Normal" and the lower half "Reversed".

Stability to partial thermal demagnetization was reasonable to 300°C, but thereafter gross instability and physical breakdown occurred.

A real reversal occurs in this sample because all tests, apart from the not entirely unexpected disintegration during acid leaching, support such a conclusion. This is the lowest reversal precisely detected in this investigation, being in the Appin Formation near the base of the Late Permian Illawarra Coal Measures. However, in reconnaissance (partially demagnetized) samples from both Wollongong 35 and Huntley 4, an apparent reversal was detected near the boundary of the Appin Formation and the underlying Tongarra coal seam.

DISCUSSION

This investigation was carried out to test further the potential for palaeomagnetic correlation in the Southern Coalfield, southern Sydney Basin. Insofar as agreement was found between results from several rock types and differing magnetic techniques, there is good evidence to support the use of palaeomagnetic correlation. Apparent reversals in Triassic samples in this investigation are probably real because Helsley and Steiner (1974) found short-term polarity zones in the Triassic Moenkopi Formation, Colorado.

There is a strong indication that near-surface weathering and precipitation of iron oxides has caused remagnetization in the sedimentary rocks down to a depth of as much as 60 m to 65 m. Such remagnetization was recorded by Schmidt and Embleton (1976) for similar rock types of Permian and Triassic age from the Perth Basin, Western Australia, and they had to reject a significant proportion of their samples. A CRM (cf. Table III) was possibly caused by secondary mineralization deposited during recent groundwater circulation. No such visual or magnetic evidence of secondary magnetization was apparent in deeper specimens. Partial AF demagnetization to 100 mT in the shallow specimens suggested that at least some remanence is carried by hematite. Results of partial chemical demagnetization do not significantly aid in the interpretation of remanence carriers. These results support the conclusion of Irving (1963, p. 2284), that the magnetization "of the cores is more stable than in surface samples" when comparing his two suites (surface and drill-hole) of Narrabeen Group samples (cf. Fig. 1). This indicates that surface samples of sedimentary rocks from the southeastern Sydney Basin may not readily provide primary palaeomagnetic directions, and further detailed work on drill cores may be necessary.

In this investigation the inclination became steeper, generally by 2° to 3° (Table III), during both partial AF demagnetization and partial thermal demagnetization. This suggests that a small component of magnetization along the present, and recent, Earth's field (inclination −65° for the southern Sydney Basin) has been removed. In the results of both Irving (1963) and Irving and Parry (1963) a similar effect can be seen.

Notwithstanding the generally good agreement between partial AF and partial thermal demagnetization results, the significant mineralogical changes during heating are sufficient to limit palaeopole determinations to AF demagnetization results. Inclusion of the thermal results would not significantly change the calculated pole positions. In Irving's (1963, table 1) data there is a slight increase in scatter after heating.

After removal of relatively soft secondary magnetizations the remanence in the sill from D2 appears to be a TRM. Sample selection avoided the widespread but highly variable alteration which affects most intrusions into the coal measures (and which prevented accurate radiometric age determination). Comparison of the inclination in the sill, and the direction in the Jurassic Mt Gibraltar syenite (Table III, cf. Boesen et al., 1961; Schmidt,

1976) suggests that the inclination in the sill is closer to the Triassic values.

The mineralogical, and extreme magnetic, changes which all grey unstained sedimentary rock specimens experienced during heating were not caused by the experimental techniques employed because several igneous rocks from another investigation were treated using the same technique without showing the marked changes. Partial thermal demagnetization and thermomagnetic studies have been carried out on other sedimentary rocks, although the results did not consistently exhibit the extreme changes found here. There are some similarities between the results for the Sydney Basin rocks and those of Dunlop (1972). It is obvious that new minerals with a high susceptibility were formed during heating of the grey and drab siltstones and sandstones. These results contrast with, for example, those of Shive and Diehl (1977) who reported work on red beds, and pointed out the importance of reduction of hematite to magnetite. It is possible that reduction in the Sydney Basin rocks is caused by trace coal (Permian and Triassic) or detrital graphite (Triassic). Reduction promoted by vacuum pump oils is not a factor in this investigation.

It is suggested that, apart from the reddened specimens, the magnetization in the sedimentary rocks in a DRM carried by detrital magnetite (perhaps hematized in part). The timing of remanence acquisition is important. Lienert and Helsley (1980) were able to confirm a reversal sequence for the Moenkopi Formation from Utah which apparently reflected detrital remanence.

Table IV shows that ricks of Permian and Triassic age from the southern Syney Basin possess very steep inclinations, thus providing data for discussion of the *Kiaman Magnetic Interval.* The top of the interval ("Reversed" to "Normal") has been termed by Irving and Parry (1963) the *Illawarra Reversal.*

McElhinny (1969, p. 63) noted that the top of the *Kiaman Magnetic Interval* occurs "within about 90 m of unsampled section between the latites" [of the Gerringong Volcanics of the Shoalhaven Group] "(showing reversed polarity) and the Lower Triassic Narrabeen sediments, which are normally magnetised". That unsampled section is at least 350 m to 400 m thick near Irving's (1963) southern sample sites (apparently in the Bald Hill Claystone). The upper 300 m of the previously-unsampled section have now been studied. Apparently the *Illawarra Reversal* is below the lowest samples of AV7. AV5 256 clearly spans a reversal — apparently "Reversed" to "Normal" (Fig. 2). The reconnaissance data from Wollongong 35 and Huntley 4 indicate an apparent reversal at a similar horizon. This indicates — but does not prove — that the *Illawarra Reversal* may occur in or just below the Appin Formation, which is (early) Late Permian in age.

The VGP for the specimens from the AV10 coal measures section is 15°N 328°E ($dp = 12°$, $dm = 13°$). Samples from six drill holes in the uppermost coal measures in this study yielded a pole position of 58°N 340°E ($dp = 09°$, $dm = 10°$). Combined partial AF and thermal demagnetization results from AV18 139 and KEM9 91 yield a pole position at 30°N 341°E. However, the

latites (basalts) studied by Irving and Parry (1963) (excluding their site 10) gave a pole position of 46°S 143°E ($dp = 13°$, $dm = 14°$). The latite pole position is almost antiparallel to that for the upper coal measures, the lowest AV10 specimens being intermediate. The Coal Cliff Sandstone pole position of 59°N 322°E ($dp = 27°$, $dm = 29°$) is close to that calculated from Irving's (1963) Narrabeen Group claystone data, which is 49°N 340°E ($dp = 14°$, $dm = 14°$) (Hicken et al., 1972, p. 32).

CONCLUSION

A 300 m composite section from nine drill holes in the Permian—Triassic rocks of the Southern Coalfield, southern Sydney Basin, has been studied palaeomagnetically. Remanence in the grey to drab specimens appears to be a DRM. Remagnetization appears to have occurred in reddened samples from depths of as much as 65 m. The palaeomagnetic results are based on both partial AF and thermal demagnetization, although mineralogical and magnetic changes during heating above 400°C are such that caution is required in the use of the latter demagnetization results.

It was hoped to locate the *Illawarra Reversal* at the top of the *Kiaman Magnetic Interval* (or Late Palaeozoic Reversed Interval), a period of reversed magnetization during the Permian. With a pole position at approximately 58°N 340°E the Permian Illawarra Coal Measures possess a pole position opposite to (antiparallel with) that of the latites (46°S 143°E) and the same as (parallel to) that of the Narrabeen Group based on this study (59°N 322°E) and previous results (49°N 340°E). Thus the "Normal" section has been extended down from the Narrabeen Group at least to near the boundary of the Appin Formation or the immediately underlying Tongarra coal seam (both Late Permian). This narrows the gap in the stratigraphic section in the "type area" of the *Kiaman Magnetic Interval* in which the *Illawarra Reversal* is apparently located to the lower half of the Illawarra Coal Measures, and it is thus (early) Late Permian. There does not appear to have been rapid movement of the magnetic pole relative to the Sydney Basin during the Late Permian and Early Triassic. The results of this investigation provide data to assist with the definition of the *Kiaman Magnetic* (or Late Palaeozoic Reversed) *Interval*, and thus for potential long-range correlation of Permian rocks and plate tectonics studies.

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